Neural integration of syntactic and visual temporal information in sentence comprehension

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During sentence processing, the brain continuously tracks the temporal structure of the unfolding speech signal. Prior work suggests that higher-level linguistic information, such as syntactic structure, provides top-down temporal predictions that help organize incoming speech (Ding et al., 2016; Coopmans et al., 2025). Conversely, visual speech, the visible articulatory movements, is thought to provide bottom-up temporal cues tied to low-level acoustic dynamics that reflect the moment-to-moment fluctuations in continuous speech (Park et al., 2016; Peelle & Sommers, 2015). However, how these two sources of temporal information are integrated remains unclear. Here, we test whether syntactic and visual timing cues contribute independently to neural encoding of audiovisual speech or interact to jointly shape neural encoding of speech.

Intracranial electroencephalography (iEEG) was recorded from seven patients as they listened to 100 Central Institute for the Deaf (CID) Everyday Sentences (Davis, 1970; Richie, Carolyn et al., 2009). Sentences were presented in five conditions: a visual-only baseline and four auditory-present conditions varying modality (auditory-alone vs. audiovisual) and acoustic degradation (temporal vs. spectral, i.e., AM- vs. FM-degraded). At the end of each session, participants were instructed to identify the correct sentence they heard in a two-alternative forced-choice task. Analyses focused on the four auditory-present conditions.

Behavioral performance was better in the audiovisual than the auditory-alone condition for both temporally and spectrally degraded speech. Consistent with prior findings (Plass et al., 2020), visual speech improved the intelligibility of degraded speech, facilitating sentence comprehension. We found no significant difference in overall speech comprehension accuracy between the AM- and FM-degraded conditions, indicating comparable task difficulty across degradation types.

For each condition, we fit multivariate temporal response function (mTRF) models (Crosse et al., 2016) to predict neural activity from acoustic predictors, with or without additional syntactic predictors. High-gamma activity (70–150 Hz) was extracted from raw iEEG data following in-house preprocessing procedures, including re-referencing and artifact rejection. All

stimulus predictors were constructed to match the resampling rate of the iEEG data. Acoustic predictors included an eight-band gammatone envelope spectrogram and an eight-band acoustic onset spectrogram. The gammatone spectrogram spanned the relevant spectral range of the stimuli to approximate cochlear transformation of the acoustic signals. The onset spectrogram was derived from the gammatone spectrogram using an auditory edge detection model (Fishbach et al., 2001). Syntactic predictors included top-down node count and word onset. The top-down parser node count was obtained from a phrase-structure parser (Kitaev & Klein, 2018). Syntax-related enhancement in neural encoding was quantified as an increase in predictive accuracy when syntactic information was added to the acoustic model.

The mTRF results showed larger syntax-related enhancement in the audiovisual compared to the auditory-alone condition. A significant interaction between modality and syntactic contribution was observed across STG electrodes. Specifically, the increase in prediction accuracy from an acoustic-only model to an acoustic + syntax model was greater for audiovisual speech than for auditory-alone speech. This pattern suggests that visual speech amplifies the contribution of syntactic information to neural encoding. Moreover, syntax-related improvements were larger for AM- than FM-degraded speech, suggesting a greater reliance on syntactic structure when temporal acoustic cues are disrupted. Together, these results indicate that syntactic and visual temporal cues interact during continuous speech processing, suggesting a cross-modal temporal scaffolding mechanism in which visual speech strengthens neural tracking of hierarchical linguistic structure.

Figure 1. Task schematic.

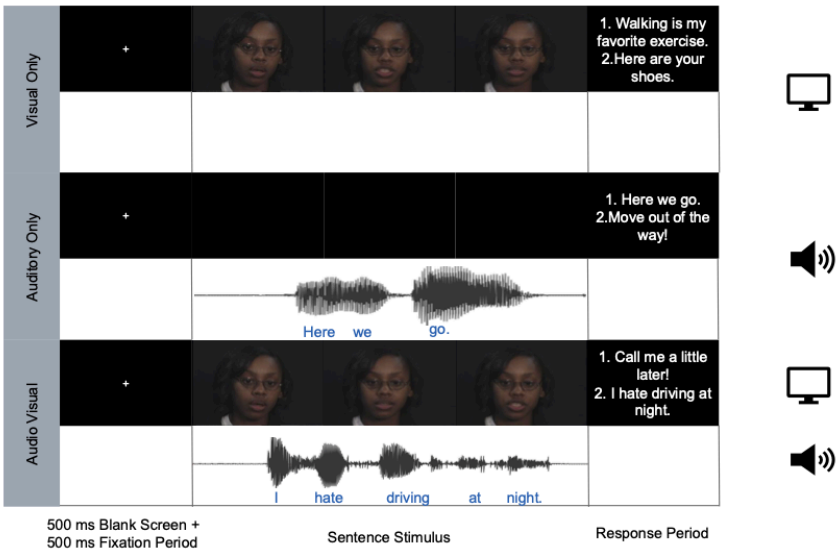
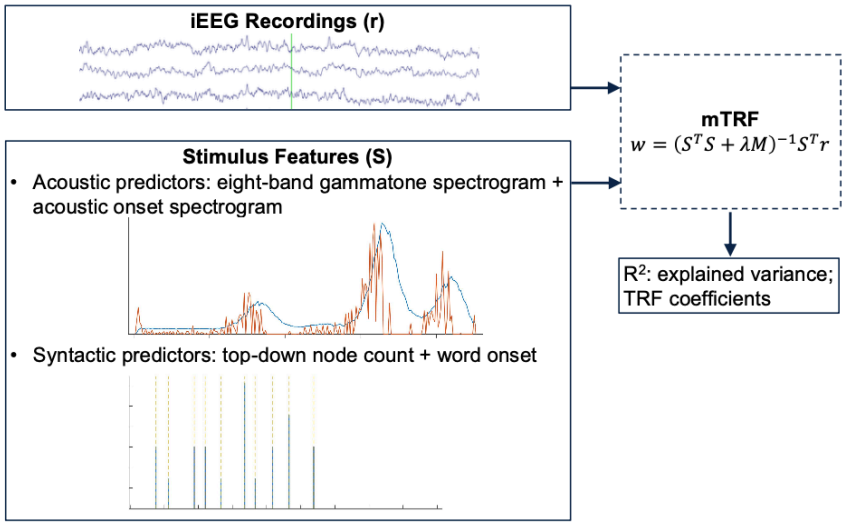


Figure 2. Analysis overview.



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